

R&D TECHNICAL REPORT

Automated ROI Re-Localization After Camera Displacement

Automatic recovery of camera-drawn monitoring regions using feature matching, robust model fitting, and edge-based refinement

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1. What's Happening

Fixed site cameras (highway, yard, plant and checkpoint cameras) are used to monitor a specific ground area, defined once as a 4-corner polygon Region of Interest (ROI) — for example, a stretch of road used for vehicle detection or counting. Cameras on poles or masts get knocked, drift, or get re-mounted after maintenance, which shifts, rotates or tilts what the camera sees. Once that happens, the original ROI coordinates no longer point at the same physical patch of ground, and the monitoring logic built on that ROI silently starts looking at the wrong area.

This project automatically re-locates a previously drawn ROI in a new frame from the same camera after it has moved, without any manual re-drawing and without needing to know how much the camera moved in advance.

2. How It's Happening — The Pipeline

Given “image 1” (the reference frame the ROI was originally drawn on) and “image 2” (a later frame from the same camera, possibly after it moved), the pipeline runs the following steps:

- Preprocess both frames — convert to grayscale, apply contrast equalization (CLAHE), and derive an edge map for each, so lighting changes and wet/dry surface differences don't confuse matching.
- Detect and match keypoints — SIFT finds distinctive points (edges, structures, markings) in both frames; matches are filtered with a ratio test, a mutual (both-directions) check, and a per-region cap so no single busy area dominates.
- Fit candidate motion models — RANSAC/MAGSAC fits three transforms to the matched points: similarity (shift + rotate + scale), affine (+ shear/stretch), and homography (+ perspective, i.e. true camera tilt/pan).
- Refine with ECC — the homography is further refined by directly aligning the two edge maps (coarse to fine), which is more precise than point matches alone.
- Score each candidate — after sanity checks (ROI must stay inside the frame, not collapse or flip), each transform is scored by how well it aligns the edges around the ROI in image 2; the best-scoring model is selected, with a bias toward the simplest model when scores are close.
- Predict the new ROI — the winning transform is applied to the original 4 ROI corners to produce their predicted location in image 2.
- Estimate confidence — the matched points are resampled 200 times (bootstrap) to see how much the predicted corners move, giving a per-corner uncertainty and an overall GOOD / CHECK verdict.
- Output — the predicted ROI, transform matrix, score and confidence are saved, and the result is visualized as the ROI drawn on image 2.

Pipeline Visualized — Test Case 1

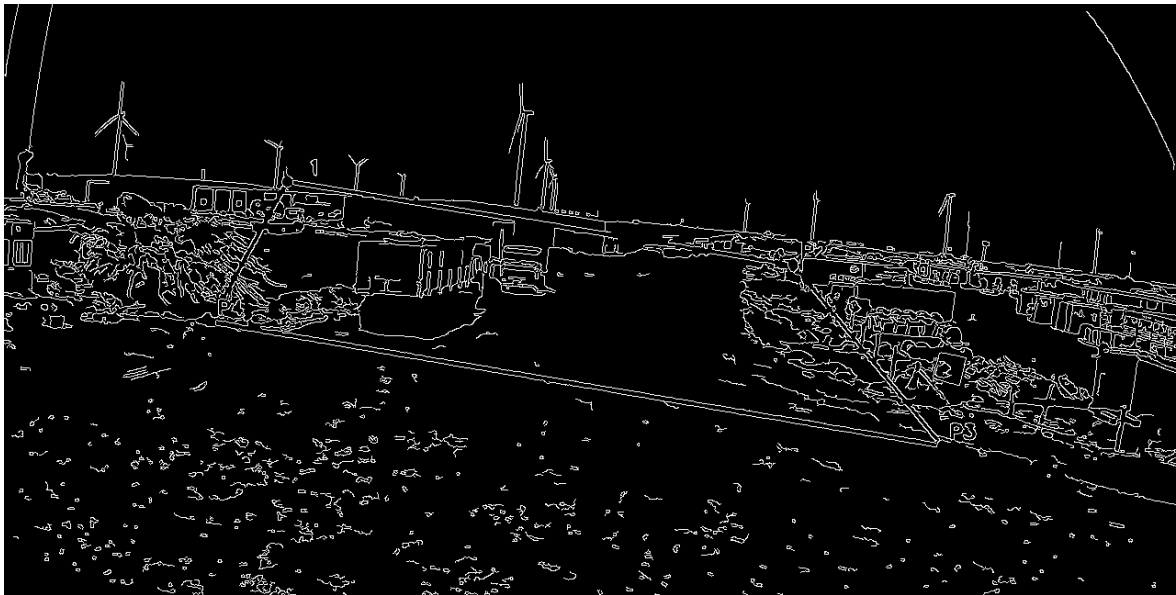
The steps above are shown here on the actual image pair from Test Case 1 (highway road segment), so each stage can be seen rather than just described.



Step 1a — Image 1 converted to grayscale



Step 1b — Contrast equalized (CLAHE), so shadows and glare don't hide structure



Step 1c — Edge map extracted from the equalized image (used later for ECC refinement and scoring)



Step 2 — SIFT keypoints detected around the ROI (each circle is a distinctive point the matcher can track)



Steps 3-4 — Matched keypoints between Image 1 and Image 2, kept only after the ratio test, mutual check and RANSAC filtering; these surviving matches are what the transform is fitted to

3. Why We're Doing This

- Eliminates manual re-calibration — someone does not need to physically revisit the camera feed and redraw the polygon every time a camera is bumped.
- Keeps monitoring accurate — detection/counting logic tied to the ROI keeps pointing at the correct ground area instead of silently degrading.
- Scales across many cameras — sites can have dozens of fixed cameras; automated recovery is far cheaper than manual review of each one.
- Works without extra hardware — no additional sensors are needed; it only needs the existing camera frames.
- Provides a confidence signal — the GOOD/CHECK verdict tells an operator when a prediction should be trusted versus manually reviewed, rather than silently accepting a bad result.

4. Results Across Different Scenarios

The pipeline was tested on multiple camera types and conditions: straight highway-style cameras, wide-angle fisheye yard cameras, hazy/overcast conditions, frames with moving vehicles in the scene, and a night-vision (infrared) camera. In every case below, the left image is the original ROI drawn on image 1, and the right image is the automatically predicted ROI on image 2.

Test Case 1 — Highway road segment, daylight



Image 1 — original ROI



Image 2 — predicted ROI

Test Case 2 — Plant road, daylight



Image 1 — original ROI



Image 2 — predicted ROI

Test Case 3 — Batching plant approach



Image 1 — original ROI

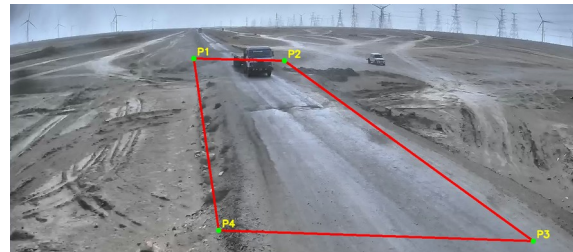
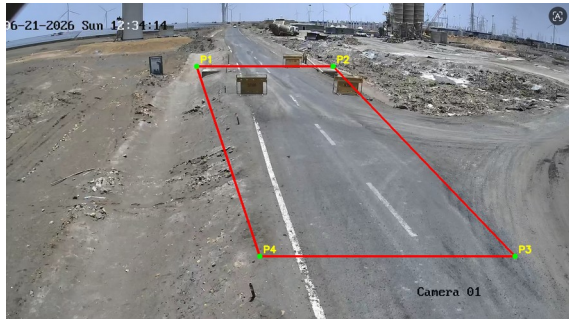
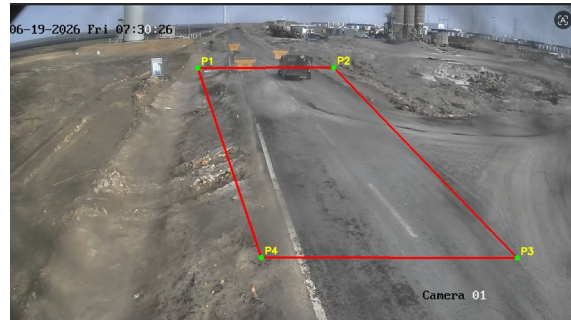
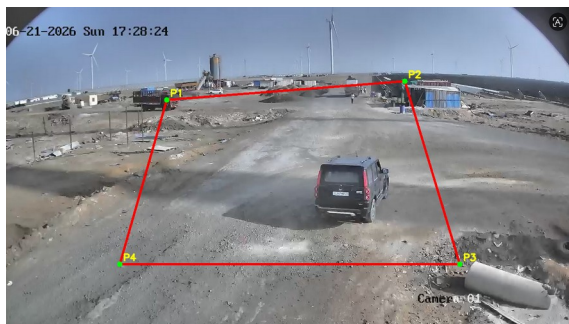


Image 2 — predicted ROI

Test Case 4 — Barrier / checkpoint road*Image 1 — original ROI**Image 2 — predicted ROI***Test Case 5 — Fisheye camera, wide road view***Image 1 — original ROI**Image 2 — predicted ROI***Test Case 6 — Fisheye camera, morning haze***Image 1 — original ROI**Image 2 — predicted ROI*

Test Case 7 — Fisheye camera, vehicle in frame

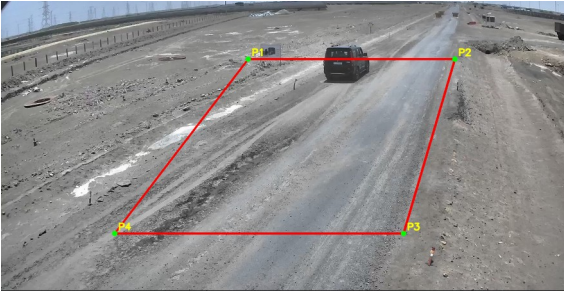


Image 1 — original ROI



Image 2 — predicted ROI

Test Case 8 — Night vision (IR) yard camera

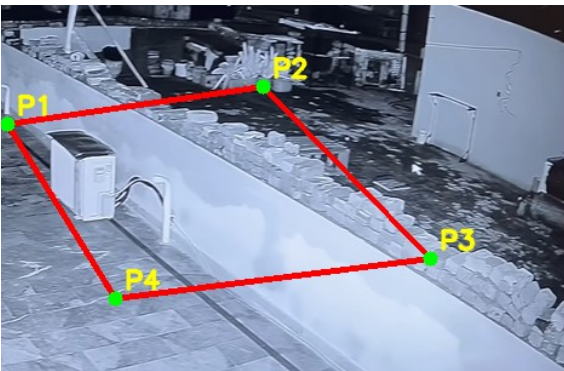


Image 1 — original ROI



Image 2 — predicted ROI (P1 off-frame, top-left — see Section 6)

5. Works on Night Vision (IR) Cameras

Test Case 8 above uses a night-vision / infrared camera feed, which has very different characteristics from daylight footage: low contrast, grain/noise, no color information, and blown-out highlights around light sources. The same pipeline — without any night-vision-specific tuning — still found enough reliable keypoints and edge structure to correctly re-locate the ROI. This is an important result because a large share of real-world camera downtime and bumping incidents happen overnight, exactly when a camera is running in IR mode.

6. Notable Result: Predicting a Corner That Isn't Even Visible

Key finding

In Test Case 8 (night vision), all four corners — P1, P2, P3, P4 — are clearly visible in Image 1, where the ROI was originally drawn. After the camera moved, corner P1's true location in Image 2 falls outside the visible frame, off the top-left edge — only P2, P3 and P4 are labelled inside the picture. The pipeline never sees any pixels at P1's new location, yet the predicted ROI edges running to P1 stay geometrically consistent with the other three corners: the line leaves the frame at the correct angle and position, exactly where P1 would sit if the frame extended further. This is only possible because the pipeline does not track each corner individually — it fits one transform from the visible matched structure and applies it to all four corners together, so a corner can be correctly placed even when it is never directly observed in the destination frame.

This matters in practice: after a camera move, part of a marked ROI can end up entirely outside the new field of view (e.g. a tilt that pushes one edge of the road off-screen). A method that could only relocate points it can directly see in both frames would fail here or simply drop that corner. Because this pipeline predicts the whole transform first and then applies it to every corner, it can correctly place corners that end up off-frame in the destination image.

7. Limitations & Open Items

Measured accuracy (synthetic + round-trip checks)

- Synthetic ground-truth check — 15 random camera-bump trials applied to a real image, each with added lighting change, blur, noise and fake moving objects: median 0.11 px, p95 0.24 px ROI-corner error. This is a best-case number (single source image, no parallax), but confirms the geometry and matching logic are sound.
- Round-trip check (image 1 → 2 → 1): mean 0.02 px, p95 0.03 px corner error, IoU 1.000 — the estimate is self-consistent. This cannot catch a shared/systematic bias, since a bias would round-trip cleanly too.

Remaining open items

- The real-image landmark check (hand-picked static points such as sign corners or pole bases, matched in both images) has not been run yet. This is the only check that measures accuracy on genuine camera movement rather than a synthetic bump, so it is the most convincing number and should be run next.
- Very low-texture or heavily obstructed scenes (e.g. dust storms, dense fog, or a scene with almost no matchable structure) could reduce the number of reliable keypoints and lower confidence.

8. Conclusion

Across highway, plant-road, checkpoint, fisheye and night-vision scenarios, the pipeline consistently re-located the ROI after simulated or real camera movement, including a case where a predicted corner ended up outside the visible frame in the destination image. Quantitatively, synthetic-bump and round-trip checks both show sub-pixel accuracy (median 0.11 px and 0.02 px respectively). This supports using it as an automated first line of defense against camera-bump drift, with the confidence score used to flag the rare cases that still need a human look — the real-image landmark check remains the next step to confirm this holds on genuine camera movement, not just synthetic bumps.